

159.735: Getting Started in CUDA

GPU Virtual Machine

Massey ITS have set up a GPU virtual machine for this course. Each student in this course has an account there. As before, replace **12345678** with your own student ID.

From a terminal window type

```
ssh 12345678@hpc-parallelgpu01
```

and log in with your usual Massey credentials. From now on, run all commands that follow from the remote host.

Try the following command.

```
nvidia-smi
```

What does the output tell you about the GPU devices available?

Sample Programs

Fetch the sample CUDA programs set up for this course:

```
wget --no-check-certificate https://moaprim.massey.ac.nz/cuda_demos.tar
```

Unpack them and go to the directory

```
tar xvf cuda_demos.tar  
cd cuda_demos
```

Compile and run the simple “hello world” program

```
nvcc -o hello hello.cu  
./hello
```

Examine the source code for **hello.cu** and see what it does. For this simple example, the kernel function doesn’t do anything.

Try running the other examples in the following order.

```
add.cu  
arradd.cu  
vecadd1.cu  
vecadd2.cu
```

Examine the code and figure out what is going on.

CUDA Device Query

Fetch a copy of the sample device query program, unpack and navigate to the directory

```
wget --no-check-certificate https://moapprime.massey.ac.nz/cuda_devq.tar
tar xvf cuda_devq.tar
cd cuda_devq
```

Compile the device query program

```
nvcc -o deviceQuery deviceQuery.cpp
```

Run the device query

```
./deviceQuery
```

What does this tell about the GPU card(s) on your system?

Lecture Material

Look at the various introductory material put on Stream. In particular note the various terminology: “host”, “device”, “kernel”, “warp”.

CUDA at Home

If you have your own Linux (Ubuntu) machine with an NVIDIA card, then installing the CUDA toolkit is easy:

```
sudo apt-update
sudo apt install nvidia-cuda-toolkit
```

This should be enough on most systems. There is also CUDA support for Windows. However, MacOSX does NOT support CUDA.